Waves

Objectives:

Students will be able to define, label and demonstrate the following vocabulary words: wave, wave motion, longitudinal wave, transverse wave, pulse, crest, energy trough, amplitude, frequency, reflection, Rarefaction, refraction and compression. Students will be able to state the properties that a medium must possess in order that waves may be transmitted through it. Students will be able to tell the essential difference between a longitudinal and a transverse wave. Students will be able to know how to make a longitudinal or compress ional pulse and a transverse pulse. Students will be able to determine the time it takes a pulse to travel down the rope or a slinky. Students will be able to understand that a wave is a means of transmitting energy.

Apparatus Needed:

Marbles, grooved ruler or stick, hose or rope, large pan of water, cork, small stone, ripple tank, medicine dropper, overhead projector, coil spring, meterstick, watch, slinky, string or fishing line, smooth pole or stiff wire, curtain rings, movies

Recommended:

1. Place five or six marbles in a groove so that they touch each other. Roll another marble against the end of the line of marbles. The vibration of waves will be transmitted through the line and the marbles on the end will roll away. Roll two, then three and so on. Observe what happens.
2. Move the hose or rope up and down to make transverse waves. Drop the stone into the water. Observe what happens.

3. Fill the flat pan about half full of water. Place the pan on an overhead projector or tabletop. Fill a medicine dropper with water. Allow one drop of water to fall into the center of the pan. Allow another drop to fall near the edge. Notice what happens in each. Sketch the pattern of waves that are seen.

4. Move a rope up and down. Describe and explain what kind of waves is observed. Cut equal-lengths of string at least one meter long. Tie a curtain ring to one end of each string. Attach the free end of a piece of string to every fifth coil of the slinky. Slip the curtain rings onto the pole [or wire], and then suspend it from the ceiling. Observe: send a longitudinal pulse down the spring by pinching together several coils of the spring and then releasing them. What kind of pulse is this? [Longitudinal or compress ional] What happens to the shape of the pulse as it travels down the wire?
Quickly pull the free end of the slinky to one side and then return it to its original position. What kind of pulse is this? What happens to the shape of the pulse as it travels down the string? Determine the time it takes a pulse to travel down the string. Change the amplitude of the pulse and again determine the travel time. How does changing the amplitude affect the speed of the wave?
Change the medium by stretching the spring to a different length. Determine the speed of a pulse in this new medium. How does changing the medium affect the speed of a wave?